

VOL. 1.

APRIL, 1888.

NO. 4.

THE HAWKEYE

ORNITHOLOGIST
AND OOLOGIST

EDITED AND PUBLISHED BY
WEBSTER AND MEAD
CRESCO, IOWA.



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*** START OF THE PROJECT GUTENBERG EBOOK THE HAWKEYE
ORNITHOLOGIST AND OOLOGIST. VOL. 1. NO. 4 APRIL 1888 ***

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R. E. RACHFORD & SON,
COLLECTING NATURALISTS
—AND WHOLESALE DEALERS IN—
BIRD SKINS AND EGGS,
BEAUMONT, TEXAS.

The Hawkeye Ornithologist and Oologist.

“Better to search the fields for health unbought,
Than see the doctor for a nauseous draught,
The wise for health on exercise depend.
God never made his work for man to mend.”

VOL. 1. CRESCO, IA., APRIL, '88.
NO. 4.

AN ECLIPSE AMONG THE ALPS.

BY H. F. HEGNER, DECORAH, IOWA.

Far along the frozen glaciers
Where the milky waters leap,
Through the fresh and quiet valleys,
Down the gorges wild and deep.

Creeps the night: The stars are shining
In the twilight and the gloom:
Drifting through the Alpine heavens,
Gently shines the rising moon.

Now she's climbing upward; shadows
Dropping from the summit's crest,
Wrap the valleys in the darkness,
Slumber on the mountain's breast.

But the earth has trailed her shadow
Far out into empty space;
And the moon walks through the shadow
With earth's image on her face.

Oh, how often has that journey
By a human soul been made;
Passing through this world of trouble,
Into sunshine—out of shade!

Long I watch her evening journey
Far above the mountain's reach:
Her sad face is almost human,
With an eloquence of speech.

For my soul is ever climbing,
Through an Alpine world of thought;

Giant BLANCS to be surmounted
Icy battles to be fought.

—*Dubuque Herald.*

OOLOGISTS VS. “RAGE” COLLECTORS.

BY W. HULL.

By a “Rage” Collector is meant one who is suddenly seized with the idea that he is deeply interested in oology, and must get together a collection of eggs as soon as possible. Some are inspired by reading oological papers, others become interested by associating with those who are at the height of excitement.

A great many persons are seized with a greater or less desire to collect stamps, coins, etc. These can be identified at any time, but eggs cannot.

A true oologist collects with a scientific purpose, actuated by a true love of nature, and an egg is of no value to him unless its identity is certain. Some collectors have a large collection, but know little or nothing about the parent birds. This would not be the case if they really cared for the eggs, but they simply keep them to gaze upon in blank admiration and boast that they have so many more eggs than some one else who may or may not collect with real interest.

I have on my tongue’s end the names of at least two dozen collectors of this class, those that collect merely for the number of eggs. These collectors can truly be called “Great American Egg Hogs.” Unrefined as this expression is, nevertheless it is to the point.

This class of collectors number many hundred throughout the United States and Canada. The excuse is “that egg collecting is a healthful and innocent pastime.” Healthful it is, if one collects the eggs himself (which is not the case with the majority) but as to the innocence, that is due to the fact that it is not taken under a full view, and as long as it is healthful and no serious results are immediately visible, it is taken for granted to be innocent. This is a matter which the American Ornithologist’s Union is acting upon, and appeals to the true oologists, for their assistance in discouraging these “naturalists”(?) in their wild career.—*Milwaukee Naturalist*.

BIRD DESTRUCTION.

BY JOS. M. WADE.

Twenty to thirty years ago, it was not an unusual sight to see even the scarlet tanager, a bright red bird with black wings and tail, flitting from tree to tree in the heart of our cities like a fiery meteor in the sun-light, and to find their nests, built very lightly of straws and similar material on the horizontal limbs of our shade trees. But they were killed or driven off long before the advent of bird millinery as a fashion. They were, indeed, a “shining mark,” and every body wanted a specimen, or thought they did, until at the present time the scarlet tanager is really a very rare bird throughout the New England States.

The Baltimore oriole, so named because the colors of the bird, black and yellow, resembled those of Lord Baltimore, has almost met the same fate, as it has done duty in ornamenting thousands of ladies’ bonnets within the past five years. Four years ago this bird was quite plenty on the elms of Boston and suburbs. The hanging nests, made of hemp, old twine, etc., were quite common. But the past season showed a great change. These birds have been shot so ruthlessly, both while here and at the South, and during the migration, that hardly a pair could be found during the breeding season of 1886.

Scientific American.

ORNITHOLOGY.

FOR THE HAWKEYE O. AND O.

THE WOOD THRUSH.

COMPOSED BY JAMES B. PURDY.

The wood thrush is singing from the depth of the glen,
His clear, bell-like music, so pleasing to me
In the fair month of May, when all nature looks gay;
They vie with each other from briar and tree.

In a deep shaded nook, where the woodbine twine,
And the dark gloomy forest conceals them from view;
By a clear, winding brooklet, o'er tangled with vines.
His dear mate is guarding her treasures of blue.

Though dark be the weather and gloomy the morn,
And all other birds in the forest are still,
And the sad face of Nature, all dreary, forlorn,
His clear, mellow notes through the dripping woods thrill.

In the evening, when nature is seeking repose,
And his dear little mate has repaired to her nest,
And the last golden sunbeams are kissing the rose,
It is then that his song is the sweetest and best.

Oh, then man why repine, be downcast on your way.
As through the long years you are journeying on;
For the sadder the morning and gloomier the day,
The happier and sweeter is the wood thrushe's song.

FOR THE HAWKEYE O. AND O.

CANADIAN FLYCATCHERS.

BY WM. L. KELLS, LISTOWEL, ONTARIO.

TRAILL'S FLYCATCHER.

(*Empidonax pusillus Trailli.*)

In size and general appearance, this species closely resembles the wood pewee; but its habitat and mode of nesting are much different. It does not frequent the back-woods nor the high timbered places; and not until a thick second-growth of low underwood succeeds the original forest in low swampy places does it make its appearance in the central districts of Ontario. Then it is so shy and wary, darting off into the deepest concealment whenever its haunts are invaded by the presence of human kind, that were it not for its noisy notes, it would scarcely be known to exist.

It arrives in this vicinity toward the end of May, when its haunts are being clothed with the emerald foliage of summer, and when it can the more easily conceal itself from observation, which it appears to dread. Then, however, the rapidly repeated "wick-we-o" of the male, as he perches on some elevated, but shady branch, intimate its presence, and that his mate has probably chosen the neighboring thicket for her summer home, while should this be penetrated, her sharp "twick," repeated in a repellant tone, gives the intruder to understand that she is there, and that his presence is not welcome.

It is very active in its movements, and darts through the shrubbery with the rapidity of a flash. It appears to subsist chiefly on insects, many of which it captures on the wing after the manner of its family. It is only in recent years that this species has become a summer resident of this vicinity; and in the particular places where it chooses to reside, it seems yearly to be more common. In the manner and position of its nest, it differs from all the other Canadian flycatchers. This is placed in deep concealment among the thick

foliage of the particular shrub, bush or underwood in which it is built, and if the first efforts at brood raising are successful, it does not appear to nest again that season, but if otherwise, it will try again. Its first nest may be found the early part of June, but its efforts at reproduction appear to cease after the month of July, and it becomes silent as August advances. 54

On the 19th of July, 1885, my boys reported to me they had found the nest of a new kind of bird in a piece of low woods on the farm opposite Wildwood. They stated that the bird was nearly as large as a hermit thrush, but more like a flycatcher; that the nest—placed in a low blue beech—was like an indigo bird's, but that the three eggs which it contained, were like those of a vireo. Eager to ascertain what this new discovery might be, I returned with the boys to the nesting place, and though the owner was absent, I saw at a glance that it was a discovery new to me. The nest was placed in the fork of a small blue beech, three feet off the ground, well concealed among the leaves and surrounding raspberry vines. It was composed externally of wool and coarse grasses; and lined with fine dry grass and some horsehair. The three eggs were of a whitish-yellow hue, with a few redish dots toward the large end. Now, anxious to see the owner of this nest, I took a ramble through the wood, where I heard and saw the male bird, and when I returned the female flushed off the nest and darted into the neighboring thicket, and for some time I supposed this species to be the olive sided flycatcher, but learning my mistake, I became certain that it was the *Trailli*, and have since been confirmed in this identification. The next summer, about the 20th day of June, within a few yards of the above mentioned place, I found in the forks of a small swamp elm, about four feet off the ground, another nest of the same species; much the same in composition, and containing three fresh eggs, similar in hue and markings. And on the same day, a few rods further in the wood, another nest of this species, containing three young a few days old. This nest, however, was in the forks of a red-maple sapling about nine feet off the ground, and some of the coarse grass stalks of which it was composed hung down nearly a foot from the bottom of the nest. Last season I failed to discover any nests of this species, though I found the birds in several other places.

THE LEAST FLYCATCHER.
(*Empidonax minimus*.)

This species, in general appearance and place of habitat, much resembles the wood pewee, but it is smaller in size, and its mode of nesting is quite different.

Its scolding notes are the repetition of a simple “chip”; but these are seldom heard except when its nest or young are approached. Its song, if such it may be called, resembles the word “chebeck” repeated in a clear tone, may often be heard, especially for some weeks after its arrival.

Its advent here usually occurs in the latter part of May; and it leaves Canada for more southern latitudes in the early part of September.

Its usual habitat is the high, rolling, hard-wood timbered lands; and for the hilly margins of gravel-bottomed creeks, it seems to have a decided partiality.

In the dry season it feeds occasionally on small fish, which it easily captures, as they wriggle in the shallow water, though in general it feeds on small insects and their progeny in various stages of development.

This little creature is quite pugilistic, and in the pairing season two males often indulge in a free and fierce fight, which probably influences the female in her decision of accepting the victor as her future partner.

The nest of this species is placed in the upright fork of a small tree, or where some small branches project from a larger stem. It is a neat, compact structure, much like that of the redstarts, composed chiefly of the fibrous matter that forms between the bark and wood of decayed trees, lined with fine hair. The set of eggs, numbering from two to five, are of a clear white hue. It does not appear to nest more than once during the season.

FOR THE HAWKEYE O. AND O.

BIRD NESTING IN THE NORTH OF ENGLAND.

BY WALTER RAINE, TORONTO, CANADA.

Crossing the stream above Thornton Force, we ascend a hill; and on reaching the top a splendid panorama opens before us. Towards the west, the river Lune wends its way for nearly twenty miles, and after passing through the town of Lancaster, empties itself into Morcambe Bay. The sun is shining on the sea, which is twenty miles away, although it does not appear half that distance. The atmosphere is very clear in this region, and taking out our field glass we plainly see several ships and steamers sailing in the Irish Sea. Towards the north are the mountain peaks of Cumberland and Westmoreland, standing out boldly; to the east stands Ingleborough mountain, towards the foot of which we make our way. We soon cross another stream which runs over Beezley Falls and down through another fairy gorge called Crina Bottom, which is similar to the one we have just traversed, with numerous cascades, and where more water ouzels, wagtails and redpolls breed; but we have not time to stay here, wishing to get on to the moors to find some plover's eggs.

Crossing some fields we find a nest of the skylark with four eggs. The mate is soaring high in the air and pouring out his joyous strains; it reminds us of the old German hymn: "Hark! Hark! the Lark at Heaven's gate sings." He is truly a wonderful songster. Immediately on leaving the ground, he begins to pour out his song, and soaring upward and upward, until he is a mere speck in the sky, when he begins to descend and does not cease singing until he reaches the earth again.

The titlark is one of the commonest birds of this district and we come across several nests, by the birds starting from in front of our feet. Their nests are always on the ground, made of dried grass, lined with hair: and contain from four to six eggs of a dusky brown, mottled over with darker brown, some having black hair lines around the egg.

A wheatear flying from out of some stones attracts our notice; and here is a nest made of grass and rabbit's fur, containing six pale blue eggs, not unlike the American blue-birds. This wheatear is called stonechat in America, though it must not be confounded with the English stonechat which is a different species.

We now reach the moors and are soon up to the knees in heather. Bird life here is very numerous. Lapwing plovers fly over our heads, crying "pewit, pewit" in a plaintive note; long-billed curlews are screaming loudly; golden plovers are whistling; red grouse are crowing; ring ouzel are calling; and above all can be heard the welcome cry of the cuckoo. High in the air, several snipes are drumming. This noise is caused by the bird's wings as it rapidly descends in the air. Nothing is more delightful than a ramble over a Yorkshire moor, where the purple heather grows in place of grass, 56 relieved here and there with stretches of bracken, gray rocks and boulders. A lapwing rises some distance in front, and marking down the spot, we soon stand gazing upon its nest and four eggs with their points inward, meeting in the center, after the fashion of all plover's eggs. The nest is simply a slight hollow in the ground, lined with bits of grass. The eggs have a dark olive ground, abundantly blotched with brown and black; average size 2.00x1.50. These eggs are much sought after as delicacies for the table; and are offered for sale in the markets at three pence and four pence each. We look around and soon find several more nests and eggs; and in less than twenty minutes we have taken some two dozen eggs, and as we don't wish to carry them along with us, we hide them beneath a rock from carrion crows until we return.

Jumping over a little brook, a snipe darts away from a tuft of grass. In the center are snugly laid four richly marked eggs of a greenish-olive hue, blotched and spotted with two or three shades of brown. A large series show the eggs to differ much in ground color and markings. This bird is numerous in Yorkshire; I have found it breeding in all parts of the county.

Only two species of duck nest in the moors, near the lagoons, the common mallard and the teal.

The mallard usually lays from seven to twelve pale olive-green tinted eggs in a nest of grass, lined with down; size about 2.25x1.60.

The teal builds a nest of vegetable substances, lined with down and feathers. Six to twelve eggs are laid of a buffy white; size, 1.75x1.25.

The short-eared owl is found nesting here; its nest is always on the ground, a simple structure of sticks, grass and heather, upon which it lays four or five white elliptical eggs, averaging in size 1.55x1.25.

Three other species of owls nest in this district: the barn owl, the tawny owl and the long-eared owl.

The latter generally selects some old crow or magpie's nest; and lays from four to six eggs, not so round as those of the short-eared species.

Tramping over the moors for a mile or so, a bird rises from the hillside before us and dashes away at a great rate. Marking down the spot, we find a stone surrounded by bird's feathers and insects' wings, and pick up a titlark which is still warm. This is the shambles of a merlin hawk who was just going to dine off the titlark when we disturbed him. We set about to look around for its nest, when my brother cried out "Here it is with four splendid eggs." In a few seconds I was there, gazing on the treasures with delight and admiration. The nest was a mixed mass of twigs, heather and brakens, raised a few inches high; and the four eggs resting in a slight hollow in the center. Their ground color was a dark, crimson brown, speckled all over with dark brown and black. Some varieties resemble eggs of the kestrel hawk, but a series of fifty eggs before me do not show such varieties in color as the eggs of the kestrel do. As a rule, they are smaller than the kestrel's eggs and not so round, nor so boldly marked. We blow the eggs and pack them away with care, and proceed farther on.

[TO BE CONTINUED.]

FOR THE HAWKEYE O. AND O.

NOTES ON SOME OF THE PASSERES OF FULTON CO., KY.

THIRD PAPER, BY L. O., PINDAR, PRES. Y. O. A., HICKMAN, KY.

Following the family *Corvidæ*, comes the family *Icteridæ*. The 57 commoner species of this family in this part of Kentucky are the red-winged blackbird, the meadow lark, the Baltimore oriole and the purple grackle. The orchard oriole and the rusty blackbird are also found; but over two years of study and careful searching in our woods and fields has failed to detect the bobolink and cowbird.

I purpose to devote this paper to the meadow lark and the Baltimore oriole.

First come; first served. The meadow or field lark is a common resident here and seems to collect in colonies. I know of two fields where I can always find them, while in other, seemingly just as favored meadows, I have failed to see them.

Early in the spring, I think, of '87, I shot at one of these birds and came very near making a clear miss as only one shot struck him and that cut off his leg. I picked him up and was going to kill him when the thought came across my mind to make a pet of him. Accordingly, on reaching home, I put him in a cage and fed him corn meal, which he ate greedily. He also relished a few wheat grains which I let him have. He grew very tame shortly, and on several occasions woke me up in the morning by his clear, rich whistling; but one day I left a lot of meal by the cage and he killed himself eating it. I would have supposed he would have known when he had enough, but he didn't.

Mr. J. B. Richards, Sec'y. Y. O. A., writes me that he has known a wild bobolink to kill itself by eating too much, and he lost a pet bobolink in the same way.

The meadow lark is accused by some writers of murdering and devouring, not only its own, but other birds' nestlings, and of being an egg-sucker; but I have nothing to offer on that point myself.

The nest of the meadow lark is made of grass, etc., built on the ground, often arched over, and sometimes at the foot of a bush or weed.

The eggs are four to six in number; crystal white, more or less marked with reddish-brown dots; average size, 1.10x.80.

And now, having called the bird a "lark" all through the article, let me say that it is not a lark at all, but a starling. The old name is, however, too firmly established to permit of a change.

The Baltimore oriole is a summer resident and quite common. The males arrive about the middle of April, and the females about ten days later. In 1887, the males arrived on the 13th, the females on the 23d. On the 30th, I found a nest nearly finished; and May 7th it contained five fresh eggs. Unfortunately the boy who tried to secure them broke the whole set.

The oriole is one of our most brilliantly plumaged birds, and on that account is much persecuted. I have heard that they stand captivity well, but as yet I have had no opportunity to try.

But it is in the nest that the chief interest in the study of this species centers. It is a closely woven pouch of various substances, grass, wool, thread, string, hair, lace, ribbons, rags, paper, leaves—all these and many more have been found in their nests. I have one by me now which is composed entirely of horsehair, with the exception of a white string around the rim to bind the hairs together; and I was shown one in Fulton, Kentucky, made entirely of long blades of grass.

I made a careful examination of the nest mentioned in the first part of the sketch. The foundation was made entirely of white wrapping twine, lined with dried grass; and as if to render it less conspicuous, it was covered with green locust leaves—it was in a locust tree—and suspended as it was, among a bunch of leaves, it was next to impossible to see it. I discovered it by watching the bird. One string in the nest was over five feet

long. It was wrapped again and again around a limb, then to the nest, woven down one side under the bottom and up the other side to the rim where it was securely fastened.

The eggs are white, with blackish spots and scrawls irregularly distributed over the surface, especially toward the larger end. The average size is 1.00x.65, according to Coues; .92x.65 according to Davie. I consider the latter more correct so far as it applies to eggs taken here.

The bird is known by the various names of fire-bird, golden robin, hang nest, etc., besides the one given above.

DEATH OF PROF. CHAS. LINDEN.

Correspondence by Ph. Heinsberger.

Prof. Charles Linden, instructor in natural history at the Buffalo High School, died in that city, of acute mania on Feb. 3.

Prof. Linden came to America from Breslau, Germany, twenty-five years ago as a sailor, and was engaged as a seaman on the great lakes when his knowledge of natural history became known and the charge of the collections of the Society of Natural Sciences was given him, after which he took his position in the High School.

The Society sent him abroad every summer, visiting in turn Brazil, where he secured many valuable birds, South America, Hayti, and the Southern States. In 1879, while exploring the coast of Labrador, he was shipwrecked and put ashore by a rescuing vessel. It was his custom to send everything new to him to the Harvard Museum. He was an authority on ornithology, and a writer of many valuable articles on that subject.

THE LARGEST OF ITS SPECIES

A HINT TO OUR GOVERNMENT.

The biggest American eagle in the United States roosts in the state house in this city. It is the property of Maj. E. J. Anderson, the state comptroller, and its roosting place is on an imitation rock in the window of his private office. The bird measures seven feet and eleven inches from tip to tip of its wings, and it is so tall that if it were alive and standing on the ground it could pick off a man's waistcoat buttons without getting on tiptoe. The profusion of little fluffy feathers on the under sides of the wings, the peculiar markings of the breast, and the depression in the top of the beak are evidences of the great age to which the bird had attained before it was killed. Those skilled in such matters estimate that it must have been at least 75 years old. It is a genuine Washington or American eagle, and probably its only rival in the country, dead or alive, is at the Smithsonian institute at Washington, but the specimen there is smaller.

The graceful yet powerful pose of this bird and the magnificent sweep of its wings show clearly how great a libel upon the bird of freedom is the atrocious figure that attempts to soar over the bundle of sticks on the back of the buzzard dollars of the present day. If the government will send an artist to Trenton it can obtain a model from which it can make a dollar that will not bring the blush of shame to the cheek of every American who has to spend it.

Maj. Anderson's eagle was shot in Hunterdon county, in this state. The bird is one of the most rare in the country, and it is scarcely ever even seen near the haunts of civilization. It is supposed that advanced age had made this bird unable to successfully pursue and capture the game of the wilderness, and that therefore it had ventured into settled regions for tamer prey. It was found near a farmer's barn, and the farmer filled it full of lead from a shotgun and a revolver without killing it, and finally captured it alive, having disabled it by wounds in the wing. It was taken to the village station, and lived there on exhibition a day or two before it died. It was then given

to Maj. Anderson, who had it stuffed and mounted, and guards it now with the tenderness and pride of a young father. He has refused for it offers running well into the hundreds of dollars.—Trenton Cor. New York Sun.

OOLOGY.

LaHoyt, Henry Co., Ia., Mch. 10, '88.

MESSRS. WEBSTER & MEAD.

Dear Sirs:—March number of the H. O. and O. at hand, and must say it is a daisy. I am an oologist, and on January last I found a nest of the great horned owl, with two fresh eggs, which is the earliest I ever knew it to breed here. The nest was in an old snag, about ten feet from the ground. The owl could be plainly seen and I could almost see the eggs while standing on the ground. I had frequently noticed her on the snag, but thought she was roosting there through the day. At the time I found the nest, the snow was on an average of two feet deep. This is the second nest of this species that I have ever found.

Red-tailed hawks are plenty here. I found eight nests in one season; they nest here in February and June, raising two broods. They always use the same nests each year unless they are disturbed. I once found a nest in which they had only laid one egg, so I went away, leaving it until they had finished the set. I visited the nest four days afterward, but that egg was gone and they never used that nest again. I have never found a nest yet of the red-tailed hawk but what it was in a tall tree, and *always leaning* over a ravine. They trouble the farmers' fowls a great deal, often killing the largest hens. I once set a steel trap by a hen which they had killed and next morning I had the male.

Yours truly,

JAMES C. JAY.



SUGGESTIONS FOR PROPERLY FORMING COLLECTIONS OF BIRDS' EGGS.

If the identification has been effected only by obtaining a good view of the birds, the fact should be stated thus: "Bird well seen," "Bird seen," or "Bd. sn.," as the case may be. For eggs not taken by the collector himself, but brought in by natives, or persons not having a knowledge of ornithology, the *local name* or the *name applied by the finder* should *only* be used, unless indeed it requires interpretation, when the scientific name may be added, but *always within* brackets thus: "Toogle-aiah (*Squatarole helvetica*);" the necessary particulars relating to the capture and identification being added. Eggs found by the collector, and *not* identified by him, but the origin of which he has reason to think he knows, may be inscribed with the common English name of the species to which he refers them; or if it has no appellation, then the scientific name may be used, but in that case *always with a note of interrogation (?)* after it, *or else* the words "Not identified." If the collector prefers it, many of these particulars may be inscribed symbolically or in short-hand, but *never unless* the system used has previously been agreed upon with persons at home, and it be known that they have a key to it. *Each specimen should bear an inscription*; those from the same nest may be inscribed; but different nests, especially of the same or nearly allied species, should never be so marked that confusion can possibly arise. It is desirable to mark temporarily with a *pencil* each egg as it is obtained; but the permanent inscription, which should always be ink, should be deferred until after the egg has been emptied. The number terminating the inscription in all cases referring to the page of the collector's note-book, wherein full details may be found, and the words or letters preceding the number serving to distinguish between different collectors, no two of whom ought to employ the same. (The initial letter of the collector's name, prefixed to the number, will often be sufficient.)

PREPARATION OF SPECIMENS.

Eggs are emptied, with the least amount of trouble, at *one* hole, which should be drilled in the *side* with an instrument called the egg-drill. The hole should, of course, be proportioned to the size of the egg and the amount of incubation it has undergone. Eggs that are hard sat upon are more easily blown by being kept a few days, but the operation must not be deferred, too long, or they are apt to burst violently immediately upon being punctured, though this may be avoided by holding them under water while the first incision is made. The hole being drilled, the lining membrane should be cleared away from the orifice with a small penknife, by which means not only is the removal of the contents, but the subsequent cleansing of the specimen facilitated. The small end of a blowpipe should then be introduced, while the other extremity is applied to the mouth and blown through, *at first very gently*. If the embryo is found to be moderately developed, a stream of water should be introduced by means of a syringe, and the egg then gently shaken, after which the blowpipe may again be resorted to, until by the ultimate use of both instruments, aided by scissors, hooks, knives and forceps, the contents are completely emptied. After this the egg should be filled with water from the syringe, shaken, and blown out, which process is to be repeated until its interior is completely cleansed, when it should be laid upon a pad of blotting paper or fine cloth, with the hole downwards, its position on the pad or cloth being occasionally changed, until it is perfectly dry. During this time it should be kept as much as possible from the light, especially from the sunshine, as the colors are then more liable to fade than at any subsequent time. In the case of very small eggs, when fresh, the contents may be sucked out by means of a bulbed tube, and the interior afterwards rinsed but as before. It is always advisable, as far as possible, to avoid wetting the outside of the shell as the action of water is apt to remove the "bloom," affect the color, and in some cases alter the crystalization of the shell. Consequently dirt stains or dung spots should never be removed. While emptying the contents, it is well to hold the egg over a basin of water, to avoid breakage in case of its slipping from the fingers. Eggs that are very hard sat upon, of whatever size they may be, should be treated in the manner detailed in "Concluding Observations," in next issue, which is a method superior to any other known at present to the writer for preventing injury arising to them. Should the yolk of the egg be dried up, a small portion of *carbonate of soda* may be introduced (but with great care that it does not touch the outer surface of the

shell, in which case the color is likely to be affected) and then the egg filled with water from the syringe, and left to stand a few hours with the hole uppermost, after which the contents are found to be soluble and are easily removed by the blowpipe, assisted by one of the hooks. It is almost unnecessary to add, except for the benefit of beginners, that the manipulation of the different instruments requires extreme caution, but a few trials will give the collector the practice necessary for success. Those who may still prefer to blow eggs by means of *two* holes are *particularly requested not to make them at the ends of the eggs, nor on opposite sides*, but on the same side. In this case the hole nearest the smaller end of the egg should be the smallest and the contents blown out at the other. If the holes are made at the *ends* of the eggs, it not only very much injures their appearance as cabinet specimens, but also prevents their exact dimensions from being ascertained accurately; and if they are made at *opposite* sides, the extent of the “show surface” is thereby lessened.

[TO BE CONTINUED.]

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THE HAWKEYE ORNITHOLOGIST & OOLOGIST

EDITED AND PUBLISHED BY
E. B. WEBSTER, F. D. MEAD.
CRESCO, IOWA.

A MONTHLY MAGAZINE
DEVOTED TO ORNITHOLOGY,
KINDRED SUBJECTS,
AND GEOLOGY.

TERMS OF SUBSCRIPTION.

Per year,	50 cents
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single copies,	5 cents

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GENERAL AGENT.—Ph. Heinsberger, 181 Ludlow St., and 89 Delancy St., New York.

All books, periodicals, specimens, etc., sent us will be reviewed.

Correspondence and items of interest relating to the several departments solicited from all. All matter for publication must be in by the last of each month in order to insure insertion in the next number.

Entered at the postoffice at Cresco, Iowa, as second class matter.

NOTES.

A neat appearing monthly devoted chiefly to philately, hails us from Mexico, N. Y., under the somewhat peculiar name of "*Common Sense*;" F. A. Thomas, editor. "May it live long and prosper."

Mr. Oliver Davie has in preparation an article on the golden-crowned thrush, which we expect to publish in a month or so. This is, in his opinion, one of his best efforts, as the bird is an especial favorite with him.

We notice that the *Bay State Oologist* appears in an improved though "condensed" form this month. Success.

Datas, 5x8, neatly printed, good paper, bound in lots of fifty, with flexible cardboard covers, 50 cents each in exchange.

The *Calmar Exchange*, published by our friend S. C. Scott, of Calmar, Iowa, comes to our table bright and fresh in its eight-page form and new dress. Those interested in the subjects of which it treats will find it an excellent paper at a small price.

Every person sending me 50 cents for one year's subscription to the HAWKEYE ORNITHOLOGIST AND OOLOGIST will receive free six varieties of rare Iowa minerals, labeled. Send postal note.

JAMES C. JAY.

LaHoyt, Henry Co., Iowa.

Among other fine specimens recently received from Messrs. Schindler & Selover, of Lake City, Minn., was one of the large, showy eggs of the Egyptian vulture, collected in Spain, which they are selling at the very low price of \$1.25 each. We can take pleasure in recommending this firm as being thoroughly reliable, and as their rates are so extremely low, we bespeak for them a most liberal patronage.

The last number of the *Collector's Illustrated Magazine*, published by E. M. Haight of Riverside, Calif., which, by the way, is our finest exchange, contains, among its many valuable articles, the first installment of a series on Vireos, by Wm. L. Kells, written in his customary interesting manner; also fine articles on "Natural History" by Chas. L. Mason; "Wonderful Mica" by J. J. Alton; and "The Cliff Dwellers of Southwestern Colorado."

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Our readers will notice that in this issue we commence the publication of a valuable article on Conchology, by J. A. Singley of Giddings, Texas. If a person has a hobby, he generally rides it well; therefore we make no hesitancy in saying that Mr. Singley's article will prove of much value and interest to the conchological student.

HOW'S THIS?

We would respectfully call the attention of the *Bay State Oologist* to the following:

Columbus, O., Nov. 17, 1888.

WEBSTER & MEAD,

Gentlemen:— * * * * *

* * I have —— copies left of the present edition (a copy of which I send you by to-day's mail) which you may have at —— per cent off. * * *

OLIVER DAVIE.

March 8, 1888.

WEBSTER & MEAD,

Gentlemen:— * * * *

I note what Mr. Foote says in regard to my "Nests and Eggs." When I sent you the —— copies they were all I had in stock—at least I thought so, for it was all I could find in the closet where I kept them. In turning over a large number of ornithological books for reference, I found a package of eight copies, one of which I sent Mr. Foote and quoted prices, but he did not take them and I sent them to Mr. Lattin who was entirely out of them and wanted them and 500 more if he could get them at once. Mr. Foote has not *purchased a single copy of the work from me. I now am entirely out until the third edition appears.* I have notified Mr. Foote to that effect. * * *

Yours truly,

OLIVER DAVIE.

Mr. Foote's article, which you have probably all seen, was decidedly ungentlemanly, to say the least.

MIGRATION NOTES.

BERNADOTTE, ILLINOIS.

BY DR. W. S. STRODE.

NAME	FIRST SEEN	NO. SEEN	NEXT SEEN	WHEN COM.
Cedar Waxwing	2-3	2		
B.-throated Bunting	2-5	7		2-13
Am. R.-legged Hawk	2-7	2		
Red-tailed "	2-7	1		
^[1] Mourning Dove	2-11	1		
Wild Geese	2-18	7		
Bluebirds	2-19	23		3-5
^[2] Brown Creeper	2-26	1		
Ducks—on river	3-1			
Robin	3-11	1		3-11
Killdeer	3-1	3		
Ducks—Old Squaw	3-3	11		
Flicker	3-3	1		
Red-w. Blackbird	3-4	12		
Meadow Lark	3-8	3		
Cooper's Hawk	3-11	2		
Pileated	3-14	2		
Woodpecker				
Purple Grackle	3-15	7		
Phoebe	3-18	1	3-19	

Am. Sheldrake	^[3] 3-18	
Sparrow Hawk	3-19	3
House Wren	3-19	2

^[1] In company with a large number of Juncas and Song Sparrows; and near some straw sheds had probably been left during the fall migration.

^[2] Found dead at foot of a tree, in very poor condition.

^[3] Numerous on the river; a friend shot three.

Great-horned Owl—found first nest February 8, containing three eggs.

Red-tailed Hawk—took first eggs, 1/2, on March 3.

Crows—counted 206 flying northwest to roost eight miles away, on March 5; on the 17th found new and finished nest.

I have made no mention of our winter residents, such as Junco, B. C. Chickadee, Tufted Titmouse, Song Sparrow, Am. Goldfinch, Crows, Jays, Hairy, Downy and Red-bellied Woodpeckers, Cardinals, etc., for scarcely a day of the period covered by this record has passed in which I have not observed more or less of all of them; and also have seen nearly every day, in my twenty to forty miles ride, more or less of some of the following more rare permanent residents: Great Horned, Barred and Screech owls, Am. Rough-legged and Red-tailed Hawks, Prairie Hens, Ruffed Grouse and Bobwhite, and occasionally a wild Turkey would cross my path.

CRESCO, IOWA.

Downy Woodpecker	2-23 1	3-4
Horned Lark	2-23 7	3-4

W.-breasted Nuthatch	3-5
Red Crossbill	3-2 1
Red-h. Woodpecker	3-3 1
Bluebird	3-15 1 3-18
Chipping Sparrow	3-15 2 3-16
Robin	3-16 2 3-17
Red-tailed Hawk	3-17 1 3-18
Pewee	3-17 1

MEDINA, N. Y.

BY N. F. POSSON.

Snow Bunting	1-1
Am. Robin	1-1
Black-c. Chickadee	1-4
White-b. Nuthatch	1-7
Brown Creeper	1-7
Downy Woodpecker	1-7
Hairy ”	1-7
Red-h. ”	1-7
Shore Lark	1-28
Gt. Northern Shrike	2-2
Bluebird	2-24
Pine Grosbeak	2-25

CHICAGO.

BY W. E. PRATT.

Am. Herring Gull	^[4] 2-4	20	2-5	2-4
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Hooded Sheldrake	2-18	4		
Red Crossbill	2-18	2		
White-r. Shrike	2-18	1		
Shore Lark	2-18	100	2-22	2-18
Bluebird	2-22	^[5] 12	2-23	
Sparrow Hawk?	2-22	4		
Red-tailed "	2-22	1		

[4] A winter resident here when the lake (Michigan) is free of ice.

[5] All males.

The Red-breasted Sheldrake and Black-capped Chickadee, winter residents, were also observed.

Those interested in bird migration will, we hope, read our "Notes" with interest. We wish to thank those who forwarded reports for their aid; and would be pleased to receive reports from all. Those wishing to aid us in this department and sending us their address will have the necessary blanks sent them.

Our home reports are meager, owing to the extreme lateness of spring.

To secure insertion, the reports should be sent not later than the 15th of each month.

CONCHOLOGY.

FOR THE HAWKEYE O. AND O.

HOW TO COLLECT AND PREPARE CONCHOLOGICAL SPECIMENS.

BY J. A. SINGLEY.

The editors having given me permission to ride my “hobby,” I intend to give brief but full instructions on the above as well as make a few remarks on matters connected with a collection. There are many collectors who can profit by what I write, but these notes are intended mainly for the class to which I belonged about 23 years ago, *i. e.*: the young collector and the beginner. Had I had these instructions then it would have saved me many a false step aside from doing some things that, while not very serious mistakes, might be called “verdant.”

I want, in the first place, to point out the advantages of collecting shells. Collecting can be done all the year round in many localities; and on the sea-shore there is no intermission. There is no climbing of trees as in egg collecting, and no danger of broken bones. Shells are not easily broken, are much handsomer than eggs; and, best of all, a shell always carries its name about with it, while you must take your correspondent’s word for the egg. And tho’ some oologists profess to be able to identify a species by the egg alone, I am bound to say that after several years of professional collecting in oology, that in the majority of cases a species can *not* be determined from the eggs.

The first thing a collector thinks of when a species of any kind is obtained is the identifying or determination. The Smithsonian Institution at Washington, D. C., or the Academy of Natural Sciences at Philadelphia, Pa., will always determine specimens sent them for that purpose. In 64 sending out specimens for naming be generous. Send all that you can spare—and perfect specimens too, as it is impossible to make a positive determination from weathered, worn or broken specimens. Don’t ask that the specimens be returned to you, but present them to the institutions or individuals to whom you sent them for naming. Besides the institutions

named above there are many of our prominent naturalists who make a specialty of conchology and will name any specimens sent to them for determination. One thing I wish to warn the beginner against is submitting his collections to an amateur and depending on his labeling. This was one of the “verdant” things of which I was guilty.

The beginner in land and freshwater shells will probably vote such a pursuit as “slow” when he first commences. After making a few exchanges and seeing the diversity of form, color, and sculpturing, he will become interested and begin making comparisons. He is then on the right road to knowledge, and as he adds species after species to his collection from land, river, or lake and sea, the hobby will grow on him and it will not be dropped when entering a business life as is the case with postage stamps and eggs, but the collection will be kept up and give many an hour’s recreation when worried with the cares of life.

Another advantage of such a collection is that you don’t offend those æsthetic people who are horrified at the idea of collecting birds and eggs and give us “fits” for “murdering” and “robbing” the poor birds. To tell the truth, after years of collecting and becoming “hardened” to it a guilty feeling sometimes comes over me when taking a set of eggs.

[TO BE CONTINUED.]

MINERALOGY.

FOR THE HAWKEYE O. AND O.

THE SCIENTIST.

BY H. F. HEGNER, DECORAH, IOWA.

But now he stands upon the sandy beach with the breakers in sight, his eyes attracted to the sea weeds and pearly shells at his feet. He is interested, and examines them carefully. Some of the shells he finds inhabited, and, as he is a naturalist, is soon acquainted with each specific form, and has a learned name for it. But he also finds a real architect in the delicate tinted coral branches at his feet. Around the head and mouth of this little creature, serving as arms for obtaining food, he finds a number of tentacles. "Nature has given you a goodly work to do, little architect," cries the naturalist, "and these tentacles are well adapted to your animal wants. *Polypus* is many armed, and henceforth, most scientifically, your name shall be polyp!"

And then he begins a pleasing study. Zoophytes he finds everywhere, spreading their beautiful architectural works along the continental borders.

He crosses the stormy Atlantic, weighing the mighty power that drives the storm. On, on through the quiet Indian ocean, the phosphorescent Indian ocean, naming and collecting myriad living forms, until he reaches the beautiful Polynesia, where, spread out in the tropical sun are the coral reefs—monuments of submerged islands—with an epitaph to the departed written in living characters around each placid lagoon.

The scientist, though, can read and understand; it is not beyond his conception; and bringing together these epitaphs, he forms a perfect image of those submerged lands. Vegetation and animalization, well defined, 65 are as clear in his imagination as though, even now, the white sunlight were reflected from those ancient islands, forming a perfect image on the retina.

He studies hard, and his conclusions, builded on the material laws of nature, are reliable; and now he returns to civilization honored and respected,

bringing the material of his researches to the civilized world.

There are other phenomena tho', fully as grand as coral islands and polyps, and he is soon in the field of work again. The gallant ship carries him through the wintry northern seas, with their ice mountains towering beneath an enfeebled sun, to the realms of perpetual snow. Past Greenland's milky glaciers that feed the Arctic main with ice mountains. Past the struggling crater of Mount Hecla, where, bound by the king of these ultimate realms, the Fire Demon struggles to be free, groaning out the essence of wrath from his fiery nostril in moulten rivers that are petrified by the rigid Ice King, and added to the adamantine chains with which he is bound; on, on to the north through a world of icebergs that moan and groan as though they were fettered in this desolate waste of frozen sea, to bar the explorer from the frosty Ice King's ultimate throne, the North Pole. But no; the scientific mind knows no defeat, and he toils on over the icy fields, while the sun, weary with his long vigil, sinks further and further in the horizon, as though he could no longer banish the sleep from his eyes, when lo!—an open Polar sea stretches away to the northward, breaking against a rocky, mountainous coast.

Filled with the joy of first discovery, the scientist voices the language of his soul in one grand apostrophe: "Oh restless Polar sea, that breaks upon this rock-bound coast, and spreads away, I know not where, e'en as Eternity, had I but my gallant ship, I'd sail thy tossing main!"

Sadly he toils back to the south, and none too soon. Creaking and roaring the massive icebergs among, on comes the tempest, and the scientist is thankful for the much needed shelter. The sun, too, has deserted him, and the grand aurora borealis, like a flaming sword above the lost Eden, seems to guard the Arctic realms, while sparkling gems glitter on each icy pinnacle.

The Arctic winter, which but for the aurora borealis would be black as the inkiest night, passes slowly away. Oh, how cold and gloomy it is! How the explorer struggles and struggles with the rigid Ice King, eagerly waiting for the departed sun to return and rescue him, and at last his anxious watch is rewarded. The east puts on the blush of modesty, a sure prophesy of his majesty's return, and immediately his welcome face appears. As he comes

up the way, the icebergs part to let him pass, and the gallant ship, freed from her rigid chains, sails onward to the south.

Thus, even thus it is that the scientist toils on and on in a masterly search for truth. Is it for glory or wealth that he dares this? No; the luxuries of civilization are even like contagion in his estimation, and with a Stoical spirit that is grand, he leads a purely intellectual life, drawing from Nature her richest treasures which she is only too glad to give. His wisdom is like a rich soil in which the seeds of knowledge and virtue germinate. He is a lover of truth, and in Nature he finds his ideal.

Natural phenomena become beautified before his studious mind, and the lower animal forms teach him objective lessons of wisdom, that, by their very simplicity, are deeply impressed on his memory.

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Even in the profound laws of chemistry and astronomy he finds a beauty that is irresistible and studies them until he develops a giant intellect. He can see beauty in truth; he can see truth in Nature; and Nature becomes his inspiration.

FOR THE HAWKEYE O. AND O.

CARE OF MINERALS.

BY W. S. BEEKMAN.

There seems to be difficulty in realizing a progress that is not in some manner dependant upon care. Care is exemplified at our several points of observation in the universe, and is realized in all things capable of advancement. Ourselves require care. Care for health, system, surroundings, character, and appearance. Among some of the many things, where, in its advancement, care greatly tends to produce a degree of perfection, equal to the amount bestowed, may be mentioned the various forms of accumulating objects for advancement. Among these forms, it will serve us at present to consider only that which has for its object-matter the accumulation of the natural chemical bodies for mineralogical study. Every one admires a prettily arranged series of rocks, and in our efforts to please both visual and intellectual senses, ever bear in mind that the direct results to be obtained are ever dependent upon the genuineness of the care bestowed.

Minerals are as much under the necessity of receiving care as is the delicate skin of an infant. Those hard and popularly considered imperishable bodies we do not exempt from the rulings of *care* as one would suppose. It is the first impulse that much govern our actions while working our specimens of the bed-rock. Specimens must be broken from their homes as carefully as one would exhume a mummy. Hammer and chisel must be deftly applied to the mother-rock, giving a nip here and a whack there, making every blow tell. After securing the specimen carefully protect all its friable or delicate parts. This can be done in many ways; often in emergencies where one does not care to utilize their handkerchiefs, a handful of grass applied to a projecting crystal will insure its safety while trimming for cabinet use.

Before trimming your specimen very much, consider carefully all the objectionable parts you wish to remove. As in a game of chess—you must know the positions and the effects of every move. Many times one will in a hurry glance over a rock and say: “Well, now, if that was trimmed about so

it would be a good thing.” Whack goes the hammer, and crumble goes the specimen. Failing to see that there was a weak place of partial fracture of the rock, which, had it been seen, could have been protected, the specimen is destroyed, owing to the lack of care. In trimming a specimen consider what you want saved and what will be better off than on. See how it can be best shaped so as to stand easily and show what is to be admired mostly. In many instances applying the chisel to the surface in directing cleavage planes will facilitate the improvement of the mineral at a less expense of battered material. A hard compact rock may be more easily broken by striking the stone while held in the hand, first protecting the hand by an old glove, than when laid on a hard surface. Where a stone is known to be quickly broken into fragments it may be found well to wrap the stone in stout paper, or cloth, before pounding. In this way the smaller fragments are easily obtained. Always trimming your specimens at the quarry your next attention should be the wrapping of each individual in soft paper, previous to being carried home.

[TO BE CONTINUED.]

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